A. AquaMoon and Two Arrays

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: brute force, greedy | https://codeforces.com/problemset/problem/1546/A

AquaMoon and Cirno are playing an interesting game with arrays. Cirno has prepared two arrays a and b, both consist of n non-negative integers. AquaMoon can perform the following operation an arbitrary number of times (possibly zero):

- She chooses two indices i and j ($1 \le i, j \le n$), then decreases the i-th element of array a by 1, and increases the j-th element of array a by 1. The resulting values at i-th and j-th index of array a are $a_i - 1$ and $a_j + 1$, respectively. Each element of array a **must be non-negative after each operation**. If $i = j$ this operation doesn't change the array a.

AquaMoon wants to make some operations to make arrays a and b equal. Two arrays a and b are considered equal if and only if $a_i = b_i$ for all $1 \le i \le n$.

Help AquaMoon to find a sequence of operations that will solve her problem or find, that it is impossible to make arrays a and b equal.

Please note, that you **don't have to minimize** the number of operations.

Input

The input consists of multiple test cases. The first line contains a single integer t ($1 \le t \le 100$) — the number of test cases.

The first line of each test case contains a single integer n ($1 \le n \le 100$).

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($0 \le a_i \le 100$). **The sum of all a_i does not exceed 100.**

The third line of each test case contains n integers $b_1, b_2, \ldots, b_n$ ($0 \le b_i \le 100$). **The sum of all b_i does not exceed 100.**

Output

For each test case print "-1" on the only line if it is impossible to make two arrays equal with some sequence of operations.

Otherwise, print an integer m ($0 \le m \le 100$) in the first line — the number of operations. Then print m lines, each line consists of two integers i and j — the indices you choose for the operation.

It can be proven that if it is possible to make two arrays equal with some sequence of operations, there exists a sequence with $m \le 100$.

If there are multiple possible solutions, you can print any.

Example

Input

```
4
4
1 2 3 4
3 1 2 4
2
1 3
2 1
1
0
0
5
4 3 2 1 0
0 1 2 3 4
```

Output

```
2
2 1
3 1
-1
0
6
1 4
1 4
1 5
1 5
2 5
2 5
```

Note

In the first example, we do the following operations:

- $i = 2, j = 1$: [1, 2, 3, 4] → [2, 1, 3, 4];
- $i = 3, j = 1$: [2, 1, 3, 4] → [3, 1, 2, 4];

In the second example, it's impossible to make two arrays equal.